Data generation machinery for N-of-1 clinical trial simulations under a two-component response decomposition: dropping the natural-history component

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1 Abstract

Background. Simulation studies of aggregated N-of-1 trials must choose where in the joint distribution a biomarker-treatment interaction lives. A companion paper compares two such choices, mean moderation and covariance moderation, under a three-component decomposition of treatment response into pharmacological (BR), placebo-belief (PB), and time-variant

natural-history (TV) components. Whether all three components are required for a comparison whose object is the architectures themselves is not obvious.

Methods. We repeat the comparison under a reduced data-generating process carrying only BR and PB, dropping the time-variant natural-history component. This is the reduction whose safety is unconditional: the implementation provides no biomarker-TV parameter, so the deleted block cannot carry any part of the biomarker coupling under any parameter setting. It is also the reduction that preserves every feature by which the trial designs differ, since the expectancy weight continues to act through PB. The reduction takes the correlation matrix from $(2+3n)$ to $(2+2n)$, from 26×26 to 18×18 at the eight occasions used throughout this program. A companion paper evaluates the alternative reduction, in which PB is dropped instead.

Results. The architecture contrast reported by paper 01 under three components is reproduced under two. At $c_{bm} = 0.30$ and $N = 70$, raising the carryover half-life from zero to one week costs covariance moderation 0.194 of power under the Hybrid design and 0.196 under OL+BDC, against 0.040 and 0.078 for mean moderation. Under the crossover design neither architecture loses power to carryover, because its off-drug occasions are spaced widely enough that residual exposure has already decayed at every one of them. At paper 01's reference cell ($c_{bm} = 0.45$, $N = 70$) all cells lie above 0.89 and the contrast is compressed by a ceiling effect to 0.032 against 0.002; the reference cell understates the architecture difference roughly sixfold and should not be used to read it. Empirical Type I error is strongly design-dependent and architecture-independent, at 0.061 under CO, 0.039 under Hybrid and 0.021 under OL+BDC against a nominal 0.05.

Conclusions. The TV component is inert for the biomarker-treatment interaction estimand under the data-generating process as implemented, and acts only as a variance term. A two-component decomposition is therefore adequate for methodological work whose object is a comparison among architectures, designs, or analysis specifications. The full three-component decomposition remains necessary when the estimand is the attribution of response to its causal sources, when a candidate biomarker may correlate with a non-pharmacological component, or when absolute power is reported as a design input rather than as a comparison.

2 1. Introduction

An aggregated N-of-1 trial pools repeated within-participant on-drug and off-drug contrasts to estimate a population-level biomarker-treatment interaction[1]. Simulation studies of such trials must specify a data-generating process, and that specification involves two choices usually made implicitly. The first is where the interaction lives in the joint distribution of biomarker and response. The second is how many distinct latent components the response trajectory is decomposed into.

The companion paper 01 takes up the first choice. It compares *mean moderation*, in which the biomarker shifts the mean of the pharmacological response, against *covariance moderation*, in which the biomarker's correlation with that response is treatment-state dependent. It shows

the two are not interchangeable: they respond differently to carryover, and the difference is large enough to change a design recommendation.

The present paper takes up the second choice. The program's standard DGP decomposes response into three components: BR, the pharmacological action of the compound; PB, the improvement driven by the participant's belief that treatment will work; and TV, drift in the underlying condition independent of treatment. That decomposition is developed and defended in paper 06, whose subject is the inferential cost of failing to decompose.

We ask what happens when the decomposition is reduced to two components, BR and TV, dropping the modelled placebo-belief term. The reduction is motivated in the first instance by tractability: it takes the correlation matrix from $(2 + 3n)$ to $(2 + 2n)$, from 26×26 to 18×18 at the eight occasions used throughout this program, and removes two of the three cross-component blocks. But the choice of *which* component to drop is not arbitrary, and Section 1.1 sets out why dropping PB rather than TV is the reduction that aligns with established practice.

2.1 1.1 Why dropping natural history is the unconditionally safe reduction

Two reductions of the three-component decomposition are available, and they are not equally safe. This paper takes the safer one; the companion paper 13 takes the other, for reasons of convention that Section 4.5 weighs against the argument given here.

The decisive consideration is structural. The biomarker cannot couple to TV at all. The implementation provides an optional biomarker-PB correlation parameter, used in paper 06's contaminated-biomarker study, but there is no biomarker-TV counterpart anywhere in the package, the drivers, or any implementation collection. The biomarker-TV entry of the correlation matrix is therefore identically zero under every parameter setting this program can express, not merely under the settings it happens to use.

That distinction has a direct consequence for what each reduction licenses. Paper 06's omitted-variable identity writes the one-component biomarker slope as $\beta_{bm}^{BR} + w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV}$, so a component contributes bias only through the biomarker's covariance with it. Deleting a block whose coupling is structurally zero cannot change the estimand. Deleting a block whose coupling is merely set to zero changes the estimand as soon as someone sets it otherwise. The present reduction is of the first kind and the companion's is of the second, which is why the companion carries a conditionality that this paper does not.

A second consideration concerns what the reduction preserves rather than what it removes. The design's expectancy weight η, which distinguishes open-label from blinded occasions, enters the data-generating process only through the PB component: it scales the PB mean trajectory and the PB standard deviation. Retaining PB therefore retains the blinding contrast, and the three trial designs continue to differ in every respect the full decomposition represents: on-drug pattern, calendar spacing, and expectancy schedule. Dropping TV costs nothing on this axis, because the calendar-time argument that drives the TV mean also drives the AR(1) week gaps of every within-component block, and those survive the reduction untouched.

Against these two points stands a consideration of convention, which is the companion paper's argument and which we do not dismiss. A time trend is a standard element of crossover and N-of-1 models, while a modelled placebo-belief trajectory is an innovation of this program rather than an inherited convention. Dropping TV therefore removes the more conventional of the two components and retains the less conventional one, which is the opposite of what a reader of that literature would expect. Section 4.5 sets the two arguments against each other rather than pretending the choice is clear-cut.

We hold everything else fixed. The trial designs, the analysis model, the parameter calibration, the biomarker-moderation levels, and the carryover half-lives are those of paper 01. Section 2 states the model and the two architectures under the reduced decomposition, Section 3 gives the results, and Section 4 takes up what follows for the design of future simulation work in this program. Three appendices carry the structural material: Appendix A on why the two architectures are treated as a pair, Appendix B on the correlation matrix the reduced DGP assembles, and Appendix C on the positive-definiteness constraint the reduction was expected to relax.

3 2. Methods

3.1 2.1 Notation and analysis model

Let $i = 1, \dots, N$ index participants and $p = 1, \dots, n$ measurement occasions at cumulative weeks $w_1 < \cdots < w_n$. Write B_i for the participant's baseline biomarker, BL_i for baseline symptom severity, and $S_{x,ip}$ for the observed symptom score. Under the two-component decomposition,

$$S_{x,ip} = \mathrm{BL}_i - (BR_{ip} + PB_{ip}),$$

so that a beneficial response lowers the score. The three-component model of paper 06 adds TV_{ip} inside the parentheses; removing it recovers the present model exactly.

Both components are drawn jointly across occasions from a multivariate normal whose means follow modified Gompertz trajectories,

$$\mathrm{mg}(t; a, d, r) = a \exp(-d \exp(-rt)),$$

evaluated at a component-specific time argument: cumulative time on drug t_{od} for BR, and accumulated belief exposure t_{pb} scaled by the design's expectancy weight η for PB. The distinct time arguments are what make the two components separable: BR responds to treatment status, PB to belief, and the blinded phases of the design move the two apart.

The analysis model is held fixed across all conditions,

$$S_{x,ip} = \beta_0 + \beta_{bm} B_i + \beta_t t_p + \beta_D D_{bc,ip} + \beta_{bm:D} B_i D_{bc,ip} + u_i + \varepsilon_{ip},$$

fitted with `nlme::lme`, a participant random intercept $u_i \sim N(0, \sigma_u^2)$, and continuous-time AR(1) residual correlation `corCAR1(form = ~ t | ptID)`. Here $D_{bc,ip}$ is the continuous exposure indicator, equal to one on drug and decaying as $\exp(-\lambda t_{sd,ip})$ off drug with $\lambda = \ln 2 / t_{1/2}$. The estimand is $\beta_{bm:D}$. Appendix A.4 states why the analysis model is held fixed.

3.2 2.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation (Architecture A)

Under mean moderation the biomarker shifts the conditional mean of the pharmacological component at on-drug occasions. After the multivariate normal draw, BR is displaced by

$$BR_{ip} \leftarrow BR_{ip} + c_{bm}\, z_i\, \sigma_{BR} \quad \text{for occasion } p \text{ on drug,}$$

where $z_i = (B_i - \mu_{bm})/\sigma_{bm}$ is the standardized biomarker. The covariance structure is untouched: the biomarker is independent of every component draw, and the interaction is carried entirely by the first moment. In the notation of Appendix B, mean moderation sets the coupling vector $b \equiv 0$.

3.2.2 2.2.2 Covariance moderation (Architecture B)

Under covariance moderation the means are untouched and the interaction is carried by a treatment-state-dependent correlation,

$$\mathrm{Cor}(B_i, BR_{ip}) = c_{bm} \cdot \phi(t_{sd,ip}), \qquad \phi(t_{sd}) = \begin{cases} 1, & \text{on drug} \\ \exp(-\lambda t_{sd}), & \text{off drug.} \end{cases}$$

The biomarker is more strongly correlated with the pharmacological response while the drug is present, and that association decays after discontinuation at the pharmacokinetic rate.

3.2.3 2.2.3 What the reduced decomposition changes

Neither architecture references TV. Removing it therefore leaves both definitions above unchanged and alters the simulation in three mechanical ways. Unlike the companion reduction, it has no substantive fourth consequence.

First, the correlation matrix shrinks, from $2 + 3n$ to $2 + 2n$. Appendix B gives the resulting block partition.

Second, the outcome loses a variance term. At the reference calibration the three components carry standard deviations of 10 (TV), 10 (PB), and 8 (BR), so TV accounts for approximately 38% of the component variance. Removing it reduces the residual variation against which the interaction is tested, and absolute power rises accordingly. This is an expected arithmetic

consequence rather than a finding, and it is why absolute power in this paper is not comparable to paper 01's. Because the TV standard deviation is not scaled by the expectancy weight, the share removed is the same under every design, which is not true of the companion reduction.

Third, the cross-component structure simplifies. The three-component matrix carries three cross-component blocks; the two-component matrix carries one. Appendix B.6 sets out exactly what is removed, and Appendix C shows that what is removed is not what constrains the moderation parameter.

What does *not* change is worth stating, because it is the principal practical difference between this reduction and the alternative. The expectancy weight η continues to act, through the PB mean and the PB standard deviation, so the open-label versus blinded contrast remains represented in the generated data and the three designs continue to differ in expectancy schedule as well as in on-drug pattern. The calendar-time argument that drove the TV mean also drives the AR(1) week gaps in every within-component block, and those are untouched, so unequal visit spacing continues to be represented as well. No feature of the trial design becomes inert.

3.3 2.3 The carryover recursion

Both architectures apply carryover to the BR mean trajectory: at an off-drug occasion, a decayed fraction of the pre-discontinuation response persists. The implementation used here differs from the one behind paper 01's published results, and the difference is worth stating.

The earlier implementation computed

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p-1}\,(1/2)^{t_{sd,p}/t_{1/2}},$$

in which $\mu_{BR,p-1}$ is the *already adjusted* value from the preceding occasion while $t_{sd,p}$ is *cumulative* time since discontinuation. Elapsed time is therefore counted twice, and every off-drug occasion after the first in an uninterrupted run is over-decayed. At a one-week half-life the second off-drug occasion receives half the residual effect it should.

The present implementation anchors the decay to the mean at discontinuation,

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p^*}\,(1/2)^{t_{sd,p}/t_{1/2}},$$

where p^* indexes the most recent on-drug occasion, so the cumulative decay factor is applied once. We verified at three reference cells (Hybrid, OL+BDC, and CO at $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$, 400 replicates under a common seed) that the correction moves power by at most 0.005 against a Monte Carlo standard error of about 0.025. It is adopted because it is correct, not because it changes results.

3.4 2.4 Simulation design

The grid crosses architecture, trial design, sample size, biomarker-moderation strength, and carryover half-life.

Factor	Levels
Architecture	mean moderation (A), covariance moderation (B)
Trial design	CO, Hybrid, OL+BDC
Sample size N	35, 70
Biomarker moderation c_{bm}	0, 0.30, 0.45
Carryover half-life $t_{1/2}$	0, 0.5, 1.0 weeks
Components	BR + PB
Replicates	500

This gives $2 \times 3 \times 2 \times 3 \times 3 = 108$ cells. The $c_{bm} = 0$ rows are the Type I error check against nominal 5%; the $t_{1/2} = 0$ rows give a no-carryover reference against which the cost of nonzero carryover is read off directly.

Throughout, N is the **total** number of participants, allocated as evenly as possible across the design's randomization paths, so a design with P paths assigns about N/P to each.

Absolute power is not comparable between architectures at the same nominal c_{bm}, since the parameter is not calibrated to equal power across the two channels. The interpretable comparison is the carryover profile within each architecture, and the qualitative shape of that profile between them.

4 3. Results

All 108 cells converged completely: the non-convergence rate is zero throughout. At 500 replicates the Monte Carlo standard error on power does not exceed 0.022, and differences smaller than about twice that should not be over-interpreted.

4.1 3.1 Power under carryover

Table 1 carries the principal result. Under the Hybrid design, moving from no carryover to a one-week half-life costs covariance moderation 0.194 of power, against 0.040 for mean moderation. Under OL+BDC the corresponding figures are 0.196 and 0.078. Both covariance-arm costs are several times the Monte Carlo standard error; both mean-arm costs are within two or three of it. This is paper 01's central finding, reproduced on a correlation matrix two-thirds the size.

The crossover design behaves differently, and consistently for both architectures: neither arm loses power to carryover under CO, and the point estimates move by less than one standard

Table 1: Power by architecture, trial design and carryover half-life at $c_{bm} = 0.30$, $N = 70$, 500 replicates per cell. Cost is the power lost in moving from no carryover to a one-week half-life. Absolute power is not comparable between architectures at a common nominal c_{bm}; the interpretable quantity is the profile across half-life within each architecture.

Architecture	Design	$t_{1/2} = 0$	0.5	1.0	Cost
Covariance (B)	CO	0.746	0.786	0.768	-0.022
Covariance (B)	Hybrid	0.878	0.778	0.684	+0.194
Covariance (B)	OL+BDC	0.746	0.610	0.550	+0.196
Mean (A)	CO	0.768	0.746	0.770	-0.002
Mean (A)	Hybrid	0.842	0.836	0.802	+0.040
Mean (A)	OL+BDC	0.754	0.692	0.676	+0.078

Table 2: The same comparison at paper 01's reference cell ($c_{bm} = 0.45$, $N = 70$). Every entry lies above 0.89, so the architecture contrast is compressed by a ceiling effect and the carryover cost is understated.

Architecture	Design	$t_{1/2} = 0$	0.5	1.0	Cost
Covariance (B)	CO	0.982	0.992	0.976	+0.006
Covariance (B)	Hybrid	0.998	0.994	0.966	+0.032
Covariance (B)	OL+BDC	0.988	0.968	0.890	+0.098
Mean (A)	CO	0.972	0.992	0.976	-0.004
Mean (A)	Hybrid	0.992	0.986	0.990	+0.002
Mean (A)	OL+BDC	0.976	0.980	0.966	+0.010

error in either direction. The reason is structural rather than statistical. CO spaces its occasions 2.5 weeks apart, so at a one-week half-life the residual exposure multiplier $\phi(t_{sd})$ has already fallen below 0.18 at the first off-drug occasion and to about 0.03 (2^{-5}) at the second. There is almost no residual exposure left for carryover to contaminate. CO's positive-definiteness ceiling is also insensitive to $t_{1/2}$ (Appendix C.4), but for a different reason: the minimum across paths is set by the placebo-first path, which has no post-exposure off-drug occasions, while the drug-first path's own ceiling rises with $t_{1/2}$. (Corrected 2026-10-08: this paragraph previously said that the 2.5-week spacing also explains the constant ceiling, and that the multiplier was below 0.03 at the second off-drug occasion. See Appendix C.4 and `analysis/report/compendium-mathematics.tex`, Section "Measurement spacing and the coupling at the first off-drug visit".)

4.2 3.2 The reference cell is ceiling-bound

Table 2 repeats the comparison at paper 01's reference cell, and is included as a caution rather than as a result. Every cell lies above 0.89. The Hybrid carryover cost under covariance moderation falls to 0.032, roughly a sixth of its value at $c_{bm} = 0.30$, not because the mechanism is weaker but because there is no room above the starting value for the loss to register.

Table 3: Empirical Type I error at the null ($c_{bm} = 0$), pooled within trial design over both architectures, both sample sizes and all three carryover half-lives (500 replicates per cell). Nominal level 0.05.

Design	cells	Rejection rate	SE	z vs 0.05
CO	12	0.061	0.0031	+3.6
Hybrid	12	0.039	0.0025	-4.5
OL+BDC	12	0.021	0.0018	-15.8

We record this explicitly because the reference cell is the natural place to look for comparability with paper 01, and doing so here would understate the architecture difference by a factor of about six. The informative cells are those where baseline power is well away from one. The same caution applies in reverse at $N = 35$, where the covariance arm's OL+BDC cost reaches 0.252.

4.3 3.3 Type I error

Type I error is not uniform across designs, and the departure is large enough to require comment. Pooled within design, the rejection rate is 0.061 under CO, 0.039 under Hybrid, and 0.021 under OL+BDC, against a nominal 0.05.

Two features of this pattern matter. It is architecture-independent: the two arms agree closely within every design, which is expected, since at $c_{bm} = 0$ the two architectures generate identical data. And it is monotone in how much off-drug information the design supplies, running from anti-conservative under CO to markedly conservative under OL+BDC.

The consequence for this paper is a restriction on what may be compared. Power figures from different designs are computed under tests with materially different actual sizes, so the cross-design power comparisons in Table 1 should be read as descriptive rather than as equal-alpha comparisons. The within-design, across-half-life comparisons that carry the paper's argument are unaffected, since the test size is held fixed along that axis.

We did not diagnose the mechanism. The companion carryover paper attributes a milder conservatism in the same analysis model to a working covariance that does not match the data-generating one, and a cluster-robust standard error removes it there. Whether the same correction would flatten the design dependence seen here is a reasonable conjecture and an untested one.

4.4 3.4 Comparison with the three-component setting

Absolute power under the two-component DGP is higher than paper 01 reports under three components, and the explanation is arithmetic. At the reference calibration PB carries a standard deviation of 10, against 10 for TV and 8 for BR, further scaled by the expectancy weight η. Removing it reduces the residual variation against which a fixed interaction is tested; on a variance argument alone, standard errors shrink and power rises correspondingly.

Because η is 0.5 at blinded occasions, the share removed is design-dependent rather than a single figure, and is largest under designs with the most open-label time.

This comparison is qualitative and should not be given a number. Paper 01's published figures were produced under the earlier carryover recursion of Section 2.3 and a partly different grid, so the difference between the two sets of power values confounds the component reduction with at least two other changes. Separating them would require a matched three-component run under the present code, which we did not perform.

What can be said without a matched run is that nothing in the *structure* of the comparison changed. The ranking of architectures within each design, the direction of the carryover effect, the insensitivity of CO, and the design ordering are all as paper 01 reports them.

5 4. Discussion

5.1 4.1 What the TV component is for

The results support a narrow claim and not a broad one.

The narrow claim is that TV is inert for the biomarker-treatment interaction estimand, and here the claim is unconditional. No biomarker-TV parameter exists anywhere in the implementation, so the coupling is structurally zero rather than set to zero, and paper 06's omitted-variable identity shows that a component uncorrelated with the biomarker cannot displace the biomarker slope regardless of its magnitude. The present results are consistent with that identity and extend it in one respect: paper 06 established it under a single architecture, and it now holds under both.

The broad claim, which we do not make, is that TV is dispensable. Three of its roles survive intact. It is required whenever the estimand is the attribution of response to pharmacology versus belief versus natural history, which is paper 06's subject. It is required whenever a candidate biomarker may correlate with natural-history drift, a case the implementation cannot currently express at all and which Section 4.3 takes up. And it contributes realism to absolute power figures calibrated against a clinical reference trajectory, so any paper reporting power as a design input rather than as a comparison should retain it.

The reduction is also, and separately, a departure from the modelling conventions of the field. Section 4.5 addresses that.

5.2 4.2 When the reduction is appropriate

We suggest the following division.

Use the two-component DGP for methodological work whose object is a comparison: between DGP architectures, between analysis specifications, between trial designs, or between inference procedures. In that setting the third component adds variance, dimension, and positive-definiteness constraints without adding anything the comparison depends on, and its absence

makes the mechanism easier to see. The saving is real but modest: the correlation matrix falls from 26×26 to 18×18, and Appendix C shows the feasible range of c_{bm} widens by only about 0.01.

Use the three-component DGP whenever a number is intended to be read as an estimate of what a real trial would achieve, or whenever the decomposition itself is the object of study.

The reduction should not be adopted for the reason that first suggests it. One might expect that deleting a third of the rows and columns would materially enlarge the feasible parameter space. Appendix C.3 shows why it does not: under compound symmetry the smallest eigenvalue of the component block is $(1 - \rho) - (c_1 - c_\times)$, which contains neither the number of occasions nor the number of components. The case for the reduction is transparency, cost, and conformity with the modelling conventions of the crossover and N-of-1 literature (Section 1.1), not numerical headroom.

5.3 4.3 A gap this paper exposes

Paper 06 states its central identity over both non-pharmacological components, and its conclusion recommends decomposition when a candidate biomarker may correlate with PB *or* *TV*. The package implements a biomarker-PB coupling and no biomarker-TV counterpart. The TV half of that claim has therefore never been exercised, and could not be without new code.

This bears on the present paper in a way that is favourable but should not be overstated. The component this paper deletes is the one whose coupling has never been exercised and cannot be, so nothing in the reduced DGP depends on the untested half of paper 06's claim, and no setting of any existing parameter can invalidate the reduction. The component it retains, PB, is the one whose coupling has been exercised and shown to matter: paper 06's contaminated-biomarker study drives the one-component bias to +0.51 at a biomarker-PB correlation of 0.45, and the present DGP can still represent that failure mode.

The overstatement to avoid is treating this as evidence that TV is unimportant. It shows only that the implementation cannot currently give TV a biomarker coupling. Under paper 06's identity a biomarker correlated with natural-history drift would displace the one-component slope exactly as a PB-correlated biomarker does, so the safety of this reduction rests on a gap in the software rather than on a property of natural history. Closing that gap, by adding the missing parameter and running the untested half of paper 06's claim, would also tell us how much the present reduction is really worth.

5.4 4.4 Limitations

The comparison inherits paper 01's restriction to a single prazosin-PTSD parameter calibration, so absolute power values should be read as illustrative and the ranking as the more portable result.

The design-dependent Type I error of Section 3.3 limits cross-design comparison, as set out there. It was not diagnosed.

The reduction removes the component that the crossover and N-of-1 literature would most expect a model to retain. A trend term is standard in that literature; a modelled belief trajectory is not. Section 4.5 weighs this against the structural argument of Section 1.1 rather than resolving it.

The comparison inherits paper 01's restriction to a single prazosin-PTSD parameter calibration, so absolute power values should be read as illustrative and the ranking as the more portable result. PTSD is, on paper 06's own taxonomy, a condition for which natural-history drift is expected to be structured and non-stationary rather than absent, since trauma anniversaries and selection at moments of clinical engagement both contribute. A reduction that deletes TV is therefore harder to defend as a clinical model of this indication than as a methodological device, and we intend it only as the latter.

The design-dependent Type I error of Section 3.3 limits cross-design comparison, as set out there. It was not diagnosed.

The reduction removes TV rather than flattening its trajectory. The two are not equivalent, since the code sets component standard deviations independently of the Gompertz maximum: a flattened TV mean leaves the variance contribution and the covariance blocks in place. A study wanting the simpler mental picture without the change in residual variance should flatten rather than remove.

No matched three-component grid was run under the present code, so Section 3.4's comparison with paper 01 remains qualitative.

The two architectures are simulated in separate replicate loops from different seeds, since the same random draw does not correspond to the same trial under different data-generating processes. Comparisons between architectures are therefore unpaired, and in any case not interpretable on an absolute scale at a common nominal c_{bm}.

5.5 4.5 Which reduction to prefer

The two papers reach the same substantive conclusion about the architecture comparison, so the choice between them is not a choice about results. It is a choice about which assumption to carry, and the considerations point in opposite directions.

The considerations divide on three axes rather than one.

5.5.0.1 Safety. No biomarker-TV parameter exists anywhere in the implementation, so the present deletion is valid under every parameter setting the software can express. The biomarker-PB coupling does exist, as an optional contamination parameter, so paper 13's reduction is valid only while that parameter is at its default of zero; a study setting it nonzero cannot use that paper's DGP at all. This axis favours the present paper.

5.5.0.2 Design structure.

The asymmetry here is one-sided. Dropping TV renders nothing inert: the calendar-time argument that drives the TV mean also drives the AR(1) week gaps of every within-component block, so unequal visit spacing survives, and the expectancy weight continues to act through PB. Dropping PB renders the expectancy weight inert, because η enters the data-generating process only through the PB mean and PB standard deviation, so the open-label versus blinded contrast disappears from the generated data. This axis also favours the present paper.

5.5.0.3 Convention.

Against those two stands the argument paper 13 makes. A time trend is a standard element of crossover and N-of-1 models, while a modelled placebo-belief trajectory is an innovation of this program rather than an inherited convention, and the placebo literature regards the belief and natural-history channels as separable only under three-arm or balanced-placebo designs that most trials do not field. The present reduction deletes the conventional component and retains the contested one. This axis favours paper 13.

5.5.0.4 Our reading.

For work internal to this program, where the contaminated-biomarker case is live and the blinding contrast is used, dropping TV is the better choice, because it is safe under every parameter setting and costs no design structure. For work intended to be read by the wider N-of-1 methodological community, dropping PB presents a more familiar model and avoids asserting a contested decomposition, at the cost of a conditionality and of an inert expectancy weight. Neither dominates, and a reader should choose by audience and by whether biomarker contamination is in scope. What should not happen is that either reduction be adopted without noticing that a choice was made.

6 5. Conclusions

1. **The architecture comparison survives the reduction.** Under a BR + PB data-generating process, covariance moderation continues to lose power as carryover grows while mean moderation does not. At $c_{bm} = 0.30$ and $N = 70$ the cost is 0.194 against 0.040 under Hybrid, and 0.196 against 0.078 under OL+BDC.

2. **TV is a variance term, unconditionally.** No biomarker-TV parameter exists in the implementation, so its coupling is structurally zero rather than set to zero, and its removal cannot change the estimand under any parameter setting. Removal raises absolute power without disturbing the ranking, the direction of the carryover effect, or the ordering among designs.

3. **The reference cell should not be used to read the contrast.** At $c_{bm} = 0.45$ and $N = 70$ every cell exceeds 0.89 and the architecture difference is compressed roughly sixfold by a ceiling effect. Comparability with paper 01's presentation is not a sufficient reason to report it as the headline.

4. **Type I error is design-dependent.** Pooled rejection rates run from 0.061 under CO to 0.021 under OL+BDC against a nominal 0.05, with the two architectures agreeing closely within each design. Cross-design power comparisons are correspondingly descriptive rather than equal-alpha.

5. **The reduction is appropriate for comparative methodological work and not for design estimates**, and it should be adopted for transparency and cost rather than for numerical headroom, which Appendix C shows it does not supply.

6. **This reduction preserves the design structure; the alternative does not.** Retaining PB retains the expectancy weight, so the open-label versus blinded contrast survives and no feature of the trial design becomes inert. The price is that the model departs from the convention of the crossover and N-of-1 literature, which carries a trend term and handles placebo by design. Paper 13 makes the opposite trade; Section 4.5 sets the two against each other.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Reproducibility

The simulation driver is `analysis/scripts/two-component-dgp/01-run-grid.R` and its output is versioned at `analysis/data/14-drop-tv-grid.rds`. The driver sets a master seed and derives a per-cell seed from it. Every number quoted in the results is read from that file at render time rather than typed in.

The two-component decomposition is supported by the `components` argument added to `generateData()` and `buildSigma()` in `R/generateData.R`. It defaults to `c("tv", "pb", "br")`, so existing calls are unaffected; passing `c("pb", "br")` produces the reduced matrix used here. The argument requires that `"br"` be retained, since BR carries the interaction channel in both architectures, and canonicalizes the ordering so that label and standard-deviation sequence is stable regardless of how the caller writes it. The package test suite passes unchanged under the default.

The corrected carryover recursion of Section 2.3 is applied in `R/generateData.R` and `implementations/tidyverse/R/functions.R`. Other implementation collections retain the earlier form; see `02-deviations.md` in the driver directory.

10 Appendix A. Why mean and covariance moderation

10.1 A.1 The two channels exhaust the Gaussian case

The simulations draw participant trajectories from a multivariate normal distribution, and a multivariate normal is completely determined by its mean vector μ and its covariance matrix Σ. Any biomarker-treatment interaction written into such a data-generating process must therefore enter through μ, through Σ, or through both. Mean moderation is the first route, covariance moderation the second, and the combined architecture developed in a companion paper[2] is the third. Within the Gaussian family there is no fourth place to put the interaction. This is why the two architectures are treated as a pair rather than as two entries in an open-ended list: they are not two options among many, but the two components of a decomposition.

Reducing the response decomposition from three components to two does not disturb this argument. The exhaustiveness claim concerns where an interaction can be written into a Gaussian joint distribution, not how many latent components that distribution is built from. Dropping TV removes rows from Σ; it does not create a third place for the interaction to live, nor remove either of the two that exist.

The exhaustiveness claim is specific to the Gaussian family and should not be read more broadly. A data-generating process built on a finite mixture, or on explicit latent responder classes, can carry an interaction in features that neither μ nor Σ captures for a single component, and Senn notes finite mixtures as an alternative variance-component model[3]. Covariance moderation is best understood as the second-moment reduced form of such a mixture, which is why the latent-class treatment is developed separately[4].

10.2 A.2 What each channel asserts

The two channels are not merely different parameterizations of one mechanism; each commits to a distinct claim about what the biomarker is.

Mean moderation asserts that the biomarker fixes a participant's drug effect. Two participants sharing a biomarker value share an expected treatment effect exactly, and any difference between their realized responses is measurement error. This is the appropriate commitment when the biomarker measures pathway engagement directly, or stands in a causal path between treatment and response.

Covariance moderation asserts that the biomarker predicts response only probabilistically. Two participants sharing a biomarker value may still respond differently, because the biomarker indexes an unobserved responder status imperfectly rather than determining response. This is the appropriate commitment when the biomarker is a statistical proxy, which is the situation the prazosin and PTSD application was designed around.

The distinction is substantive rather than technical. It determines what a fitted interaction coefficient means, and how much power carryover costs.

10.3 A.3 Standing in the literature

Mean moderation is the prevailing convention wherever biomarker-treatment interactions are simulated. Haller and Ulm[5] place the interaction in a regression mean, as do the oncology design literatures on enrichment and predictive-marker trials[6,7]. Grenet and colleagues[8] use the same form for a crossover. That literature is, however, overwhelmingly parallel-group.

The N-of-1 methodological literature does not adopt mean moderation, because it does not model an observed moderator at all. Its standard formulation carries a random participant-by-treatment interaction, a variance component describing unexplained heterogeneity with no covariate attached[3,9,10]. Biomarker-validation simulation for N-of-1 designs is therefore close to unoccupied territory rather than a settled field with a convention to follow, and Hendrickson et al.[1] is its principal entry. The two-channel decomposition of A.1 is adopted precisely because no established N-of-1 convention exists to inherit.

10.4 A.4 Holding the analysis model fixed

Both architectures are analyzed with the identical model of Section 2.1. This is deliberate and worth stating explicitly, because a data-generating process and an analysis model are separate objects, and a comparison that varies both attributes differences to neither. The continuous exposure variable `Dbc` loses contrast variance as carryover grows, and a binary on-drug indicator does not. A comparison in which one architecture carried a continuous regressor and the other a binary one would confound that effect with the architecture contrast it was meant to isolate. Holding the analysis model fixed means every difference reported in Section 3 is attributable to the data-generating process alone.

The same argument governs the reduction studied here. The analysis model does not reference the component decomposition at all: it sees only S_x, and S_x is a sum. Removing TV therefore changes what the analysis is fitted to without changing what it fits, which is the property that makes the comparison in Section 3 interpretable.

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11 Appendix B. Correlation structure under the two-component decomposition

This appendix sets out the correlation matrix the reduced data-generating process assembles, in the notation of paper 01's Appendix B so that the two can be read side by side. Symbols

follow the variable names used in the code: BM is the biomarker, written B_i in Section 2.1, and BR_p, PB_p the response components at occasion p.

Mean moderation appears below only through the statement that its coupling vector is zero. It writes no biomarker-response correlation into the covariance at all; its interaction is carried in the mean instead, as Section 2.2.1 sets out.

11.1 B.1 Variable ordering and block partition

Each participant's trajectory is drawn from a single multivariate normal over an ordered variable list. With n measurement occasions at cumulative weeks $w_1 < \cdots < w_n$, the two-component vector is

$$X = (\,\mathrm{BM},\ \mathrm{BL},\ PB_1, \ldots, PB_n,\ BR_1, \ldots, BR_n\,)^\top, \qquad \dim X = 2 + 2n,$$

against $2 + 3n$ under the three-component decomposition. The ordering is component-major: all n occasions of one component are contiguous, rather than all components at one occasion. Writing A_c for the $n \times n$ within-component block of component $c \in \{pb, br\}$ and $C_{pb,br}$ for the single cross-component block, the correlation matrix partitions as

$$R = \begin{pmatrix} 1 & 0 & 0^\top & b^\top \\ 0 & 1 & 0^\top & 0^\top \\ 0 & 0 & A_{pb} & C_{pb,br} \\ b & 0 & C_{br,pb} & A_{br} \end{pmatrix}.$$

Three features carry over unchanged from the three-component case. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker BM couples only to the response component BR, through the vector $b \in \mathbb{R}^n$; its correlation with PB is zero throughout this paper, by the default setting of the optional contamination parameter rather than by structural necessity. And b is the only place in R where the architectures differ.

The third feature deserves emphasis because it is what licenses the reduction. In the three-component matrix the biomarker's row is $(1, 0, 0^\top, 0^\top, b^\top)$: zero against both TV and PB. Deleting the TV rows and columns therefore deletes only zeros from that row, and the coupling vector is untouched, entry for entry.

Here the argument requires no qualification. The TV entries are zero because the implementation provides no parameter that could make them otherwise, so the reduction is valid under every setting the software can express. The companion paper's reduction deletes the PB block instead, whose entries are zero by a default rather than by construction, and its Appendix B.1 records the corresponding caveat.

11.2 B.2 Within-component blocks A_c

Let ρ_c denote the autocorrelation parameter of component c (`c.pb`, `c.br` in the code). The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks $4, 8, 9, 10, 11, 12, 16, 20$, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

We note that the code uses the cumulative week gap, not the occasion index difference, and that this choice is consequential: Appendix C.4 reports that the index-based form would cost roughly a fifth of the feasible range of c_{bm}.

11.3 B.3 The cross-component block $C_{pb,br}$

Let c_1 denote the same-occasion cross-component correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-component correlation (`c.cfct`). The same-occasion entry is a constant and the different-occasion entries inherit the AR(1) decay:

$$[C_{pb,br}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times \, \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

Under the three-component decomposition there are three such blocks, $C_{tv,pb}$, $C_{tv,br}$ and $C_{pb,br}$, all of this form. The reduction retains the last and deletes the first two.

11.4 B.4 Biomarker coupling vector b

The vector b carries the biomarker-treatment interaction under covariance moderation. Its entry is gated on the drug indicator and decays with time since discontinuation $t_{sd,p}$ at rate $\lambda = \ln 2 / t_{1/2}$:

$$b_p = \begin{cases} c_{bm}, & \text{occasion } p \text{ on drug} \\ c_{bm}\, e^{-\lambda t_{sd,p}}, & \text{occasion } p \text{ off drug.} \end{cases}$$

Under mean moderation $b \equiv 0$ and the interaction is carried by the displacement of μ_{BR} described in Section 2.2.1. The two architectures are therefore distinguished, in the language of this appendix, by exactly one object: whether b is populated or zero.

11.5 B.5 A worked example

To make the blocks concrete, take four occasions drawn from the Hybrid design at cumulative weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_\times = 0.1$, $c_{bm} = 0.45$, and a carryover half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$(|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

The unequal spacing matters: occasions one to three are a week apart, while the fourth sits four weeks after the third.

11.5.1 B.5.1 Within-component block A_{br}

With $\rho_{br} = 0.6$,

$$A_{br} = \begin{pmatrix} 1 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1 \end{pmatrix}.$$

The fourth occasion is nearly uncorrelated with the first three, a direct consequence of the four-week gap: $0.6^4 = 0.1296$ against $0.6^1 = 0.6$ for an adjacent pair.

11.5.2 B.5.2 Cross-component block $C_{pb,br}$

With $c_1 = 0.3$, $c_\times = 0.1$ and the decay taken at $\rho = 0.6$,

$$C_{pb,br} = \begin{pmatrix} 0.3000 & 0.0600 & 0.0360 & 0.0047 \\ 0.0600 & 0.3000 & 0.0600 & 0.0078 \\ 0.0360 & 0.0600 & 0.3000 & 0.0130 \\ 0.0047 & 0.0078 & 0.0130 & 0.3000 \end{pmatrix}.$$

The block is symmetric here because the two components share a decay rate in this example; in general it need not be.

11.5.3 B.5.3 Biomarker vector b

Occasions one and four are on drug, occasions two and three off it with $t_{sd} = 1$ and 2. With $c_{bm} = 0.45$ and $\lambda = \ln 2$,

$$b = (0.4500,\ 0.2250,\ 0.1125,\ 0.4500)^{\top}.$$

The off-drug entries halve with each additional week since discontinuation, and the fourth occasion returns to full strength because the participant is back on drug. It is this alternation that carries the interaction, and its erosion under carryover that Section 3 measures.

11.6 B.6 What the reduction removes, and what it does not

Relative to the three-component matrix, the reduction removes:

- one within-component block, A_{tv} ($n \times n$);
- two cross-component blocks, $C_{tv,pb}$ and $C_{tv,br}$;
- n rows and columns from R, taking it from $2 + 3n$ to $2 + 2n$: at $n = 8$, from 26×26 to 18×18.

It removes no entry of b, because the biomarker never couples to TV under any setting. It removes no parameter from the architecture definitions of Section 2.2, because neither references TV. And it renders nothing inert: the expectancy weight η continues to act through the PB component, and the calendar-time argument that drove the TV mean also drives the AR(1) week gaps, which survive (Section 2.2.3).

One distinction is worth recording, because it is easy to conflate two operations that are not equivalent. Removing the component is not the same as flattening its trajectory. The code sets component standard deviations independently of the Gompertz maximum, so setting that maximum to zero yields a flat TV mean while leaving the TV variance contribution, its block A_{tv}, and its cross-component blocks in place. A study wanting the simpler mental picture without the change in residual variance should flatten; a study wanting the smaller matrix should remove. This paper removes.

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12 Appendix C. Positive definiteness and the feasible range of c_{bm}

The correlation matrix of Appendix B must be positive definite for the multivariate normal draw to be valid. It is not automatically so: the moderation parameter c_{bm} is bounded above, and the bound depends on the correlation structure, the trial design, and the number of occasions. This appendix derives the bound, gives a closed form for the compound-symmetric case, and reports where the reduction of Appendix B does and does not relax it.

The motivation is practical. Simulation drivers in this program repair a non-positive-definite matrix silently, by projection onto the nearest positive-definite matrix, so a grid cell that violates the bound does not fail. It quietly becomes a different data-generating process, with a realized biomarker-response correlation no longer equal to the nominal c_{bm}. Knowing the bound in advance is therefore a precondition for trusting a grid.

12.1 C.1 The constraint is a Schur complement on the biomarker row

Partition the correlation matrix of Appendix B.1 by separating the biomarker from everything else. Because BL is uncorrelated with all other variables, it contributes an identity block that factors out of every determinant and may be dropped. What remains is

$$R' = \begin{pmatrix} 1 & \tilde{b}^\top \\ \tilde{b} & M \end{pmatrix}, \qquad M = \begin{pmatrix} A_{pb} & C_{pb,br} \\ C_{br,pb} & A_{br} \end{pmatrix},$$

where M is the $Kn \times Kn$ component block ($K = 2$ here, $K = 3$ under the full decomposition) and $\tilde{b} \in \mathbb{R}^{Kn}$ is the biomarker coupling extended by zeros: it equals b on the BR positions and vanishes elsewhere.

By the Schur complement, R' is positive definite if and only if M is positive definite and

$$1 - \tilde{b}^\top M^{-1} \tilde{b} > 0.$$

Writing $\tilde{b} = c_{bm} v$, where v is the unit-scaled coupling pattern with $v_p = \phi(t_{sd,p})$ on the BR positions and zero elsewhere, the condition becomes $c_{bm}^2 \, v^\top M^{-1} v < 1$, giving the exact ceiling

$$c_{bm}^* \;=\; \left(v^\top M^{-1} v\right)^{-1/2}. \tag{1}$$

Three consequences follow immediately, and they organize the rest of this appendix. The ceiling depends on the component block only through the quadratic form $v^\top M^{-1} v$, so components that are weakly coupled to BR contribute little to it. It depends on the design through v, which encodes the on-drug and off-drug pattern. And it is unaffected by mean moderation, under which $b \equiv 0$ and the constraint reduces to positive definiteness of M alone.

12.2 C.2 Closed form for the component block under compound symmetry

When the within-component blocks are compound symmetric rather than AR(1), M admits an exact spectral decomposition. This is the structure of the published implementation discussed in paper 01, and it is worth recording because it explains a result of C.3 that is otherwise surprising.

Under compound symmetry with a common within-component correlation ρ, common same-occasion cross-component correlation c_1, and common different-occasion cross-component correlation $c_\times$, the component block is a Kronecker sum,

$$M \;=\; A \otimes I_n \;+\; B \otimes (J_n - I_n), \tag{2}$$

where J_n is the $n \times n$ matrix of ones and A, B are $K \times K$:

$$A = I_K + c_1(J_K - I_K), \qquad B = \rho I_K + c_\times(J_K - I_K).$$

A describes the same-occasion structure across components, B the across-occasion structure. Since $J_n - I_n$ has eigenvalues $n-1$ once and -1 with multiplicity $n-1$, and I_n commutes with it, the Kn eigenvalues of M collapse to those of two $K \times K$ reductions:

$$\operatorname{eig}(M) \;=\; \underbrace{\operatorname{eig}(A + (n-1)B)}_{\text{once each}} \;\cup\; \underbrace{\operatorname{eig}(A - B)}_{\text{multiplicity } n-1}.$$

Both reductions are of the form $\alpha I_K + \beta(J_K - I_K)$, whose eigenvalues are $\alpha + (K-1)\beta$ once and $\alpha - \beta$ with multiplicity $K-1$. Substituting gives all four distinct eigenvalues in closed form:

$$\begin{array}{ll}
\text{all-ones time direction:} & [1 + (n-1)\rho] + (K-1)[c_1 + (n-1)c_\times], \\
 & [1 + (n-1)\rho] - [c_1 + (n-1)c_\times] \quad (K-1 \text{ times}), \\
\text{time-contrast directions:} & (1-\rho) + (K-1)(c_1 - c_\times), \\
 & (1-\rho) - (c_1 - c_\times) \quad (K-1 \text{ times, each with multiplicity } n-1).
\end{array}$$

At the program's reference values $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, the three-component block has exactly four distinct eigenvalues, 7.7, 5.0, 0.5 and 0.2, the last with multiplicity $2(n-1) = 14$.

12.3 C.3 Why the component count barely matters

The smallest eigenvalue in C.2 is generally the last,

$$\lambda_{\min}(M) = (1-\rho) - (c_1 - c_\times), \tag{3}$$

and the striking feature of (3) is what does not appear in it. It contains neither n nor K. Under compound symmetry the positive-definiteness of the component block is independent of both the number of occasions and the number of components. Adding a third component does not make M harder to keep positive definite; nor does adding occasions.

This is the analytic explanation for a result that would otherwise be puzzling. One might expect that deleting a third of the rows and columns from a correlation matrix would materially enlarge the feasible parameter space. It does not, and (3) shows why: the binding structural constraint never depended on the count in the first place.

Two further readings of (3) are worth recording. The operative cross-component quantity is the *difference* $c_1 - c_\times$, not either coefficient separately, so keeping the same-occasion and across-occasion couplings close to one another matters more than keeping either small. And positive definiteness requires $\rho < 1 - (c_1 - c_\times)$, which at the reference values is $\rho < 0.9$: a compound-symmetric structure cannot represent very high autocorrelation at all.

Under AR(1) the corresponding statement is different, and no equally compact closed form is available, because A_c is Toeplitz rather than exchangeable and does not commute with the cross-component structure in the way (2) requires. Numerically, the AR(1) component block becomes progressively more ill-conditioned as occasions are added, since each new occasion is strongly correlated with its neighbours. The number of *occasions* therefore does constrain c_{bm} under AR(1), while the number of *components* still does not.

12.4 C.4 Numerical ceilings

Evaluating (1) directly at the program's reference parameters ($\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, AR(1) in cumulative weeks), minimized across each design's randomization paths:

Design	$t_{1/2}$	Three components	Two components
CO	0.0	0.619	0.627
CO	0.5	0.619	0.627
CO	1.0	0.619	0.627
Hybrid	0.0	0.456	0.469
Hybrid	0.5	0.506	0.516
Hybrid	1.0	0.544	0.552
OL+BDC	0.0	0.452	0.464
OL+BDC	0.5	0.500	0.509
OL+BDC	1.0	0.533	0.540

The reduction buys between 0.007 and 0.013 of feasible range (the smallest gain is OL+BDC at $t_{1/2} = 1$, $0.540 - 0.533$; corrected 2026-10-08 from 0.008). This is consistent with C.3: the deleted blocks were coupled to BR only through c_1 and $c_\times$, both small, so they contributed little to the quadratic form $v^\top M^{-1} v$ that sets the ceiling.

Three further observations follow from evaluating (1) under variations of the structure, none of which is specific to the reduction.

First, the ceiling rises with the carryover half-life within the two non-crossover designs, from 0.456 to 0.544 under Hybrid. A longer half-life makes the coupling vector v smoother, and a smoother v aligns better with the well-conditioned directions of M. The crossover design's ceiling is insensitive to $t_{1/2}$ because the minimum across paths is set by the placebo-first path, whose off-drug occasions all precede exposure and so carry zero coupling at every half-life (no drug has yet been given, so there is nothing to carry over). The drug-first path's own ceiling does rise with the half-life, from 0.619 at $t_{1/2} = 0$ to 0.623, 0.639 and 0.651 at 0.5, 1 and 2 weeks under the three-component structure. (Corrected 2026-10-08: this sentence previously attributed the constancy to the 2.5-week spacing of the off-drug occasions. See `analysis/report/compendium-mathematics.tex`, Section "Measurement spacing and the coupling at the first off-drug visit", and `analysis/scripts/compendium-mathematics/check-crossover-binding-path.R`.)

Second, the number of occasions matters considerably more than the number of components. Holding the design shape fixed and varying n alone, the ceiling falls from approximately 0.81 at $n = 4$ to 0.58 at $n = 12$ under a three-component structure. That is an order of magnitude larger than the effect of deleting an entire component.

Third, the exponent scale in A_c is consequential. Using the occasion-index difference $|i - j|$ in place of the cumulative week gap $|w_i - w_j|$, an otherwise identical AR(1) structure, lowers the worst-cell ceiling from approximately 0.45 to 0.35. The code uses the week gap; documentation elsewhere in this project has described it as the index difference, and the two are not interchangeable.

12.5 C.5 Practical implications

For the present paper, (1) confirms that the grid of Section 2.4 is feasible throughout: the largest moderation level used is $c_{bm} = 0.45$, against a worst-cell ceiling of 0.464 under the two-component structure. The margin is real but not generous, and a study wishing to extend the grid past $c_{bm} = 0.45$ should evaluate (1) first rather than rely on the simulation to complain, since it will not.

More generally, the appendix identifies which levers move the feasible range and which do not. The number of latent components does not, and neither does the choice between two and three of them. The number of measurement occasions does, substantially. The within-component correlation ρ does, non-monotonically, with the reference value 0.7 close to the maximum of the ceiling. And the cross-component coefficients matter through their difference rather than their magnitudes.

12.5.0.1 Provenance. The Kronecker identity (2) and the eigenvalue decomposition of C.2 were verified numerically against direct construction for $K \in \{2, 3\}$ and $n \in \{4, 8, 12\}$. The ceilings in C.4 were computed from (1) and agree with an independent grid search over

c_{bm} in steps of 0.01 to within the grid step. The observations in C.4 concerning occasion count, exponent scale, and ρ were obtained by the same grid search under a reimplemented matrix builder rather than the production `buildSigma`, and should be read as characterizing the structure rather than as production audit results.

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